



Practical 4.

WAP to Implement 3D Object Using
JAVAFX

```
import javafx.application.Application;  
import javafx.scene.*;  
import javafx.scene.paint.Color;  
import javafx.scene.paint.PhongMaterial;  
import javafx.scene.shape.Box;  
import javafx.scene.shape.Cylinder;  
import javafx.scene.shape.Sphere;  
import javafx.scene.transform.Rotate;  
import javafx.stage.Stage;
```

```
public class ThreeDObjects extends  
    Application {
```

@Override

```
public void start(Stage primaryStage){
```

```
    Box box = new Box(100, 100, 100);
```

```
    PhongMaterial boxMaterial = new  
        PhongMaterial(Color.BLUE);
```

```
    box.setMaterial(boxMaterial);
```

```
    box.setTranslateX(150);
```

```
    box.setTranslateY(150);
```

```
box.setTranslateZ(400);  
Sphere sphere = new Sphere(50);  
PhongMaterial sphereMaterial =  
    new PhongMaterial(Color.ORANGE);  
sphere.setMaterial(sphereMaterial);  
sphere.setTranslateX(350);  
sphere.setTranslateY(150);  
sphere.setTranslateZ(400);  
Cylinder cylinder = new Cylinder(40, 100);  
PhongMaterial cylinderMaterial =  
    new PhongMaterial(Color.  
        PALETOLETTRED);  
cylinder.setMaterial(cylinderMaterial);  
cylinder.setTranslateX(250);  
cylinder.setTranslateY(300);  
cylinder.setTranslateZ(400);  
PointLight pointLight = new PointLight(  
    Color.WHITE);  
pointLight.setTranslateX(250);  
pointLight.setTranslateY(100);  
pointLight.setTranslateZ(200);
```




```
AmbientLight ambientLight = new  
    AmbientLight (Color.rgb(100,100,100));
```

```
Group root = new Group(box, sphere,  
    cylinder, pointLight,  
    ambientLight);
```

```
PerspectiveCamera camera = new  
    PerspectiveCamera(true);
```

```
camera.setTranslateZ(-1000);  
camera.setNearClip(0.1);  
camera.setFarClip(2000.0);  
camera.setFieldOfView(35);
```

```
Scene scene = new Scene(root, 600, 400, true);  
scene.setFill(Color.GRAY);  
scene.setCamera(camera);
```

```
Rotate rotateX = new Rotate(20,  
    Rotate.X_AXIS);
```

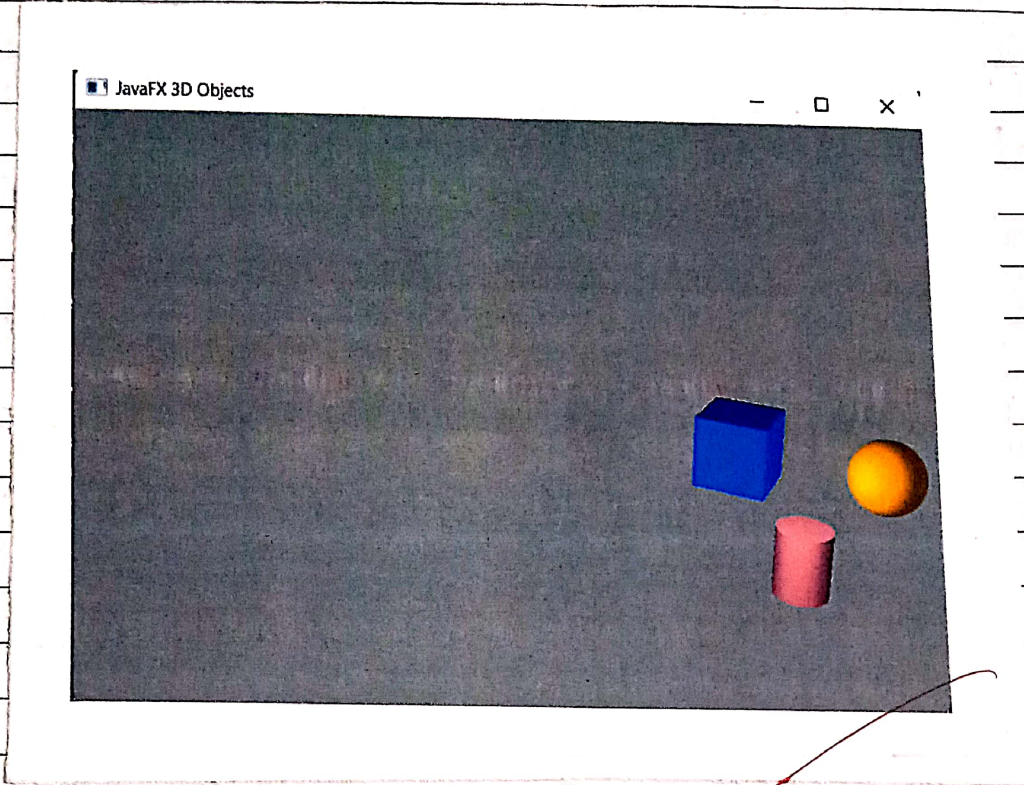
```
Rotate rotateY = new Rotate(30,  
    Rotate.Y_AXIS);
```

```
root.getTransforms().addAll(rotateX,  
    rotateY);
```

```
primaryStage.setTitle("JavaFx 3D Objects");  
primaryStage.setScene(scene);  
primaryStage.show();
```



```
public static void main (String[] args) {  
    launch(args);  
}
```



Output

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